

RANDNLA DICE

Greetings, fellow science dweller! Before you lies a curious manuscript describing the RandNLA dice - a clever (if we do say so ourselves) artifact that playfully links an object beloved by gamers and gamblers alike - the humble die - with our cherished field of Randomized Numerical Linear Algebra (RandNLA).

This field, once the exclusive domain of theoretical computer science, is now making its way to become a powerful ally of the high-performance computing professionals in the battle against the waning magic of moore's law. As computational demands outpace traditional methods, RandNLA offers speed, scalability, and just enough chaos to keep things interesting. and what better symbol for structured randomness than a pair of dice?

This document serves as a decoder for the cryptic symbols etched onto the RandNLA dice, revealing two essential categories of knowledge: select flagship algorithms - those that currently lead the charge in scalable randomized computation; and the core ideas - foundational principles that echo across nearly every method in the RandNLA landscape. We hope you'll enjoy exploring the RandNLA dice as much as we delighted in designing them. may your next sketch be small in size and in distortion- with high probability!

whp

With High Probability

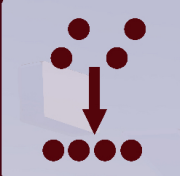
A technical term meaning that the probability of failure decays rapidly with increasing dimension, often exponentially or faster. In RandNLA, this guarantees that randomized methods yield accurate results almost surely as the problem size grows. This concept underpins the robustness of randomness-driven algorithms in both theory and practice.

Σ

RSVD

Randomized Singular Value Decomposition justifiably holds the top position among popular RandNLA methods. Over the years, it has played a central role in popularizing the field, with implementations widely available in software packages such as MATLAB, cuSOLVER, scikit-learn, and others. RSVD approximates the dominant singular values and vectors of a large matrix by projecting it onto a lower-dimensional random subspace (found using a sketching operator), significantly reducing computational cost compared to classical deterministic methods while retaining accuracy with high probability.

Ref: "Finding Structure with Randomness," N. Halko et al. (2011)



Johnson-Lindenstrauss Lemma

A foundational result stating that a set of high-dimensional points can be embedded into a much lower-dimensional space via a random linear map, with pairwise distances approximately preserved. This lemma provides the theoretical backbone for dimensionality reduction in RandNLA, justifying the use of random projections in compressing large-scale data with minimal distortion.



Blendenpik

A fast solver for overdetermined least-squares problems, Blendenpik leverages the "sketch-and-precondition" paradigm: it first constructs a preconditioner from a sketch via a QR factorization, then solves the preconditioned system with a Krylov subspace method such as LSQR or conjugate gradients. Its design effectively supercharges classical solvers like those in LAPACK by transforming the problem into a well-conditioned one, achieving both high performance and numerical stability.

Ref: "Blendenpik: Supercharging LAPACK's Least-Squares Solver," H. Avron et al. (2010)



Subspace Embedding

A generalization of the Johnson-Lindenstrauss lemma to entire subspaces. A random map is said to be a subspace embedding if it preserves the geometry (lengths and angles) of all vectors in a fixed low-dimensional subspace. This concept is central to the theoretical guarantees for many RandNLA methods.

K

RBKI

Randomized Block Krylov Iteration is an algorithm that constructs a Krylov subspace using randomized sketches to improve low-rank approximations (e.g., SVD). Each iteration adds spectral information, making it particularly effective for matrices with slowly decaying spectra, where basic RSVD tends to struggle. Although Krylov methods with random starting vectors have long been known, their theoretical and practical behavior is only recently becoming understood through modern random matrix theory.

Ref: "Randomized algorithms for low-rank matrix approximation: Design, analysis, and applications," J. Tropp et al. (2023)

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Sketching Operators

Random matrices or transforms used to compress large datasets into smaller, manageable "sketches." These sketches preserve essential numerical properties of the original data, making them critical tools in RandNLA. Popular sketching techniques include Gaussian sketches, sparse transforms, and subsampled randomized Hadamard transforms. The choice of sketch and tuning parameters can significantly affect both the performance of the resulting algorithm.

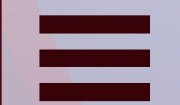
Ref: "Sketching as a Tool for Numerical Linear Algebra," D. Woodruff (2014)

[CUR]

RCUR

CUR is a low-rank matrix approximation method that, unlike SVD, retains interpretability by using actual rows and columns from the original matrix. This makes it especially useful in data science and scientific computing, where preserving data meaning is critical. Rather than relying on full-rank decompositions, RCUR uses randomized techniques to efficiently select informative rows and columns. Most RCUR implementations are based on randomized column subset selection or interpolative decompositions, both of which relate closely to randomized QR with column pivoting (QRCP), discussed next.

Ref: "Relative-Error CUR Matrix Decompositions," P. Drineas et al. (2008)



Coherence

Measures how concentrated the energy (or "information") of a matrix is across its rows or columns. Low coherence implies uniform spread, ideal for random sampling, while high coherence indicates localized structure, making naive random sampling unreliable. Coherence is often treated as the "condition number" of sampling, playing a vital role in the success of randomized matrix approximations. However, its relevance depends on the choice of sketching technique: for example, the behavior of Gaussian sketching is entirely independent of coherence.

Ref: "Randomized Algorithms for Matrices and Data," M. W. Mahoney (2011)



BQRRP

Blocked QR with Randomization and Pivoting is a modern framework for constructing efficient QRCP algorithms using suitable subroutines. Classical QRCP has long resisted formulation in terms of Level-3 BLAS operations, limiting its performance on modern architectures. BQRRP overcomes this by enabling high-performance, in-place QR factorizations applicable to general matrices. BQRRP can outperform traditional QRCP by up to two orders of magnitude on CPUs and achieve up to 65% of the performance of unpivoted QR on CPUs and GPUs. This framework is a successor to the HQRRP method, introduced by Martinsson et al. in 2025.

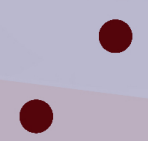
Ref: "Anatomy of High-Performance Column-Pivoted QR Decomposition," M. Melnichenko et al. (2025)



Tuning Parameters

Randomized algorithms depend on hyperparameters like oversampling factors, block sizes, and iteration counts. These parameters balance runtime efficiency, accuracy, and success probability. Theoretical analysis can be used to choose these parameters in a way that leads to rigorous guarantees on algorithm performance. In many settings (i.e., combinations of algorithms and datasets), it is safe to set parameters to achieve far lower computational cost than theoretical analysis would predict.

Ref: "Surrogate-Based Autotuning for Randomized Sketching Algorithms in Regression Problems," Y. Cho et al. (2025)



RandRAND

Randomized Range Deflation is a modern class of randomized preconditioners for solving large-scale linear systems. Rather than computing explicit eigenpairs or low-rank approximations, RandRAND deflates the difficult portion of the spectrum via projection onto a random subspace. Techniques such as Q-less randomized QR or iterative orthogonalization are used to isolate dominant eigenmodes, thus reducing the condition number and accelerating Krylov subspace solvers. The method is numerically stable and provides condition number bounds that depend only weakly on problem size, often collapsing to a constant if the deflation subspace matches the effective spectral dimension.

Ref: "Preconditioning via Randomized Range Deflation (RandRAND)," O. Balabanov et al. (2025)